

3.1.2 How the body works

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 4			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Respiration releases the energy needed for living processes and is represented by the equation: glucose + oxygen → carbon dioxide + water (+ energy) Students should know the word equation for respiration. Students should know that glucose is derived from the diet and that carbon dioxide and oxygen gases are exchanged through the lungs.	Details of aerobic/anaerobic respiration are not needed.	4.4.2.1	4.2.1.1
Lifestyle can have an effect on people's health eg diet and exercise are linked to obesity; smoking to cancer; alcohol to liver and brain function. A healthy diet contains the right balance of the different food groups you need and the right amount of energy.	Suggested activity for TDA Compare the energy released by burning a 'low fat' crisp (or rice cake) with a normal one. No knowledge of deficiency diseases is required. Knowledge of the effects of an unbalanced diet is limited to a person being underweight or overweight and includes the link with Type 2 diabetes.	4.2.2.5 and 4.2.2.6	4.3.1.2
People who exercise regularly are usually fitter than people who take little exercise.	The concept of fitness is limited to the rate at which the pulse rate returns to normal for a person after exercise. Suggested activity for TDA Investigate the effect of exercise on pulse rate. Suggested activity for TDA Investigate the effect of caffeine drinks on pulse rate.	4.2.2.6	4.3.1.2